

Basic Vulnerability Assessment Using OpenVAS (GVM)

Objective

The objective of this task was to perform a vulnerability scan on a personal computer using a standard security tool and identify potential security weaknesses. This task helps in understanding how vulnerabilities are detected, analyzed, and mitigated in real-world systems.

Tool Used

Greenbone Vulnerability Management (OpenVAS)

System Details

- Operating System: Kali Linux
- Target IP Address: 127.0.0.1 (Localhost)
- Scan Type: Full and Fast
- Scan Duration: Approximately 30–60 minutes

Methodology

1. Installed and configured OpenVAS (GVM) on Kali Linux
2. Initialized vulnerability feeds and database
3. Accessed web interface via browser
4. Created a target using localhost (127.0.0.1)
5. Configured a scan task using “Full and Fast” profile
6. Executed the vulnerability scan
7. Analyzed the scan results
8. Documented identified vulnerabilities and mitigation steps

OPENVAS

Security Information >

Configuration >

Targets

Port Lists

Credentials

Scan Configs

Alerts

Schedules

Report Configs

Report Formats

Scanners

Filters

Tags

Administration >

Help >

New Target

X

Name

my kali scan

Comment

Hosts

☒ Manual 127.0.0.1

☐ From file

Exclude Hosts

☒ Manual

☐ From file

Allow simultaneous scanning via multiple IPs

☒ Yes ☐ No

Cancel

Save

OPENVAS

Dashboards

Scans >

Tasks

Reports

Results

Vulnerabilities

Notes

Overrides

Assets >

Resilience >

Security Information >

Configuration >

Administration >

Help >

New Task

X

Name

full vulnerability

Comment

Scan Targets

My Kali Scan

Alerts

Schedule

--

☐ Once

Add results to Assets

☒ Yes ☐ No

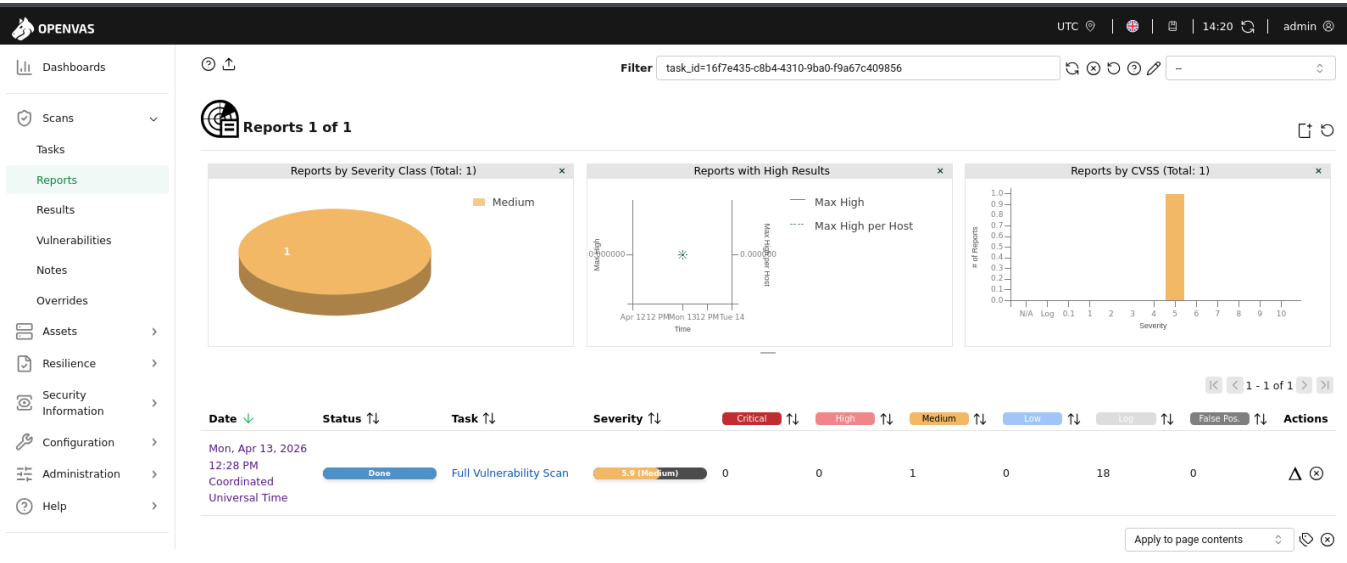
Apply Overrides

Cancel

Save

Scan Summary

- Total Vulnerabilities Detected: 19
- Informational (Log): 18
- Medium Severity: 1
- High Severity: 0
- Critical Severity: 0



Identified Vulnerability

Vulnerability Name

SSL/TLS Medium Strength Cipher Suites Supported

Severity Level

Medium

CVSS Score

5.9

Description

The scan identified that the system supports SSL/TLS cipher suites that are classified as medium strength. These cipher suites may not provide strong encryption and could allow attackers to compromise secure communication.

OPENVAS							
UTC 14:34 admin							
Dashboards							
Scans							
Tasks							
Reports							
Results							
Vulnerabilities							
Notes							
Overrides							
Assets							
Resilience							
Security Information							
Configuration							
Administration							
Help							
1 - 10 of 18							
Name	Oldest Result	Newest Result	Severity	QoD	Results	Hosts	
SSL/TLS: Report Weak Cipher Suites	Mon, Apr 13, 2026 12:31 PM Coordinated Universal Time	Mon, Apr 13, 2026 12:31 PM Coordinated Universal Time	5.9 (Medium)	98 %	1	1	
Services	Mon, Apr 13, 2026 12:29 PM Coordinated Universal Time	Mon, Apr 13, 2026 12:29 PM Coordinated Universal Time	0.0 (Log)	80 %	2	1	
SSL/TLS: Report Non Weak Cipher Suites	Mon, Apr 13, 2026 12:31 PM Coordinated Universal Time	Mon, Apr 13, 2026 12:31 PM Coordinated Universal Time	0.0 (Log)	98 %	1	1	
SSL/TLS: Collect and Report Certificate Details	Mon, Apr 13, 2026 12:30 PM Coordinated Universal Time	Mon, Apr 13, 2026 12:30 PM Coordinated Universal Time	0.0 (Log)	98 %	1	1	
SSL/TLS: Report Perfect Forward Secrecy (PFS) Cipher Suites	Mon, Apr 13, 2026 12:31 PM Coordinated Universal Time	Mon, Apr 13, 2026 12:31 PM Coordinated Universal Time	0.0 (Log)	98 %	1	1	
SSL/TLS: Version Detection	Mon, Apr 13, 2026 12:30 PM Coordinated Universal Time	Mon, Apr 13, 2026 12:30 PM Coordinated Universal Time	0.0 (Log)	80 %	1	1	
OS Detection Consolidation and Reporting	Mon, Apr 13, 2026 12:30 PM Coordinated Universal Time	Mon, Apr 13, 2026 12:30 PM Coordinated Universal Time	0.0 (Log)	80 %	1	1	
Unknown OS and Service Banner Reporting	Mon, Apr 13, 2026 12:31 PM Coordinated Universal Time	Mon, Apr 13, 2026 12:31 PM Coordinated Universal Time	0.0 (Log)	80 %	1	1	
Hostname Determination Reporting	Mon, Apr 13, 2026 12:38 PM Coordinated Universal Time	Mon, Apr 13, 2026 12:38 PM Coordinated Universal Time	0.0 (Log)	80 %	1	1	

Technical Explanation

SSL/TLS protocols are used to secure communication over networks. Cipher suites define how encryption is performed. Medium-strength cipher suites use weaker encryption algorithms compared to modern standards, making them less secure.

Impact

- Possible interception of encrypted communication
- Increased risk of Man-in-the-Middle attacks
- Potential exposure of sensitive data

Affected Component

- SSL/TLS configuration of local service

Risk Assessment

Although the vulnerability is not critical, it poses a moderate risk and should be addressed to improve overall system security.

Recommended Mitigation

- Disable medium-strength cipher suites
- Enable strong encryption algorithms such as AES
- Use TLS 1.2 or TLS 1.3
- Regularly update SSL/TLS configurations
- Apply latest system and security updates

The screenshot shows the OpenVAS web interface. On the left is a sidebar with navigation links: Dashboards, Scans, Assets, Resilience, Security Information, NVTs (highlighted), CVEs, CPEs, CERT-Bund Advisories, DFN-CERT Advisories, Configuration, Administration, and Help. The main content area displays the report for 'NVT: SSL/TLS: Report Medium Cipher Suites'. It includes tabs for Information, Preferences, and User Tags. The 'Information' tab is active, showing a 'Summary' section with a description: 'This routine reports all Medium SSL/TLS cipher suites accepted by a service.' Below this is the 'Scoring' section, which includes 'CVSS' details: CVSS Base (0.0 Log), CVSS Base Vector (AV:N/AC:L/Au:N/C:N/I:N/A:N), CVSS Origin (N/A), and CVSS Date (Wed, Mar 7, 2012 8:44 AM Coordinated Universal Time). The 'Insight' section states: 'Any cipher suite considered to be secure for only the next 10 years is considered as medium.' The 'Detection Method' section shows 'Quality of Detection: remote_app (98%)'. The 'Family' section is 'SSL and TLS'.

OPENVAS

Dashboards

Scans

Assets

Resilience

Security Information

NVTs

CVEs

CPEs

CERT-Bund Advisories

DFN-CERT Advisories

Configuration

Administration

Help

NVT: SSL/TLS: Report Medium Cipher Suites

Information Preferences (0) User Tags (0)

Summary

This routine reports all Medium SSL/TLS cipher suites accepted by a service.

Scoring

CVSS

CVSS Base 0.0 (Log)

CVSS Base Vector AV:N/AC:L/Au:N/C:N/I:N/A:N

CVSS Origin N/A

CVSS Date Wed, Mar 7, 2012 8:44 AM Coordinated Universal Time

Insight

Any cipher suite considered to be secure for only the next 10 years is considered as medium.

Detection Method

Quality of Detection: remote_app (98%)

Family

SSL and TLS

Informational Findings

The majority of findings were informational and included:

- CPE Inventory detection
- Operating system identification
- Service detection
- SSL certificate details
- Hostname reporting

These findings do not represent direct vulnerabilities but provide useful system insights.

Challenges Faced

- Initial feed update timeout during setup
- SSL certificate warning in browser
- Understanding scan configurations and output

Outcome

- Successfully performed a vulnerability scan using OpenVAS
- Identified and analyzed system vulnerabilities
- Gained practical knowledge of vulnerability assessment
- Understood real-world security risks and mitigation techniques

Conclusion

This task provided hands-on experience with vulnerability scanning tools and techniques. The system was found to be mostly secure, with only one medium-level vulnerability related to SSL/TLS configuration. Proper mitigation steps can further strengthen the system's security posture.